

# Enterprise Data Governance & Data Quality Audit Report

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## 1. Executive Summary

This document presents an end-to-end Data Governance and Data Quality audit conducted on the Enterprise Sales Dataset. The project involved transforming unmodeled flat-file data into a structured **Star Schema Data Model**, defining clear metadata classifications, establishing a standardized **Business Glossary**, and deploying an automated **Data Quality Dashboard** in Power BI.

Through this implementation, the Overall Data Health Index reached **86.5%**, successfully identifying critical data ingestion errors across departments for targeted remediation.

## 2. Business Glossary

The Business Glossary provides standardized, enterprise-wide definitions for key business entities and metric calculations to resolve ambiguity across business units.

| Term          | Business Definition                                                                          | Data Owner      | Technical Rule         |
|---------------|----------------------------------------------------------------------------------------------|-----------------|------------------------|
| Gross Revenue | The total gross value of a sales transaction prior to any discounts or deductions.           | Sales & Finance | Gross Revenue >= 0     |
| Net Revenue   | The net monetary amount realized from a sale after applying allowable promotional discounts. | Finance         | Net = Gross - Discount |

| Term                   | Business Definition                                                                             | Data Owner      | Technical Rule         |
|------------------------|-------------------------------------------------------------------------------------------------|-----------------|------------------------|
| <b>Active Customer</b> | A registered customer entity that has performed at least one valid registration or transaction. | Marketing & CRM | Valid cust_id required |

### 3. Technical Metadata & PII Classification

To ensure compliance with data protection standards, the data model has been decentralized into a Star Schema, accompanied by technical metadata and **Personally Identifiable Information (PII)** security classifications.

**Table 1: Dim\_Customer (Dimension Table)**

- **Description:** Stores master customer profiles and identity data.

| Column Name | Data Type | Key Type        | Security / PII Level      | Description                              |
|-------------|-----------|-----------------|---------------------------|------------------------------------------|
| cust_id     | String    | Primary Key (1) | Public                    | Unique customer identification code.     |
| full_name   | String    | Attribute       | Confidential (PII)        | Full legal name of the customer.         |
| email       | String    | Attribute       | Confidential (PII)        | Primary electronic contact address.      |
| birth_date  | Date      | Attribute       | Highly Confidential (PII) | Date of birth for demographic profiling. |

## Table 2: Dim\_Department (Dimension Table)

- **Description:** Contains organizational units responsible for data asset ownership.

| Column Name | Data Type | Key Type        | Security / PII Level | Description                            |
|-------------|-----------|-----------------|----------------------|----------------------------------------|
| dept_id     | Integer   | Primary Key (1) | Public               | Surrogate primary key for departments. |
| dept_owner  | String    | Natural Key     | Public               | Organizational department name.        |

## Table 3: Fact\_Sales (Fact Table)

- **Description:** Records transactional revenue metrics and associated quality flags.

| Column Name   | Data Type | Key Type           | Security / PII Level | Description                               |
|---------------|-----------|--------------------|----------------------|-------------------------------------------|
| cust_id       | String    | Foreign Key (Many) | Internal             | Relational key linking to Dim_Customer.   |
| dept_owner    | String    | Foreign Key (Many) | Internal             | Relational key linking to Dim_Department. |
| gross_revenue | Currency  | Measure            | Restricted           | Raw transactional gross amount.           |
| discount_amt  | Currency  | Measure            | Restricted           | Calculated or applied discount amount.    |

| Column Name        | Data Type     | Key Type        | Security / PII Level | Description                                     |
|--------------------|---------------|-----------------|----------------------|-------------------------------------------------|
| net_revenue        | Currency      | Measure         | Restricted           | Transaction net realization value.              |
| Is_Complete_Name   | Integer (0/1) | Validation Flag | Internal             | Data quality indicator for name presence.       |
| Is_Valid_Email     | Integer (0/1) | Validation Flag | Internal             | Quality flag for standardized email syntax.     |
| Is_Valid_Revenue   | Integer (0/1) | Validation Flag | Internal             | Quality flag ensuring non-negative gross value. |
| Is_Valid_net_Logic | Integer (0/1) | Validation Flag | Internal             | Quality flag verifying business rule logic.     |

## 4. Data Quality Audit Log & Remediation Strategy

An exhaustive audit of 1,500 records revealed key operational issues affecting data integrity:

### Identified Issues Log

#### 1. Whitespace & Empty String Anomaly (Completeness):

- **Issue:** Blank entries encoded as empty text ("") rather than formal SQL nulls, causing false completeness evaluation.
- **Remediation:** Applied Power Query Trim transformations and replaced empty strings with proper null values.

#### 2. Syntax Corruption in Email Formatting (Validity):

- **Issue:** Email strings containing structural syntax errors (e.g., \_at\_ instead of @).

- **Remediation:** Flags raised via conditional column parsing; bad records routed for automated CRM data cleaning.

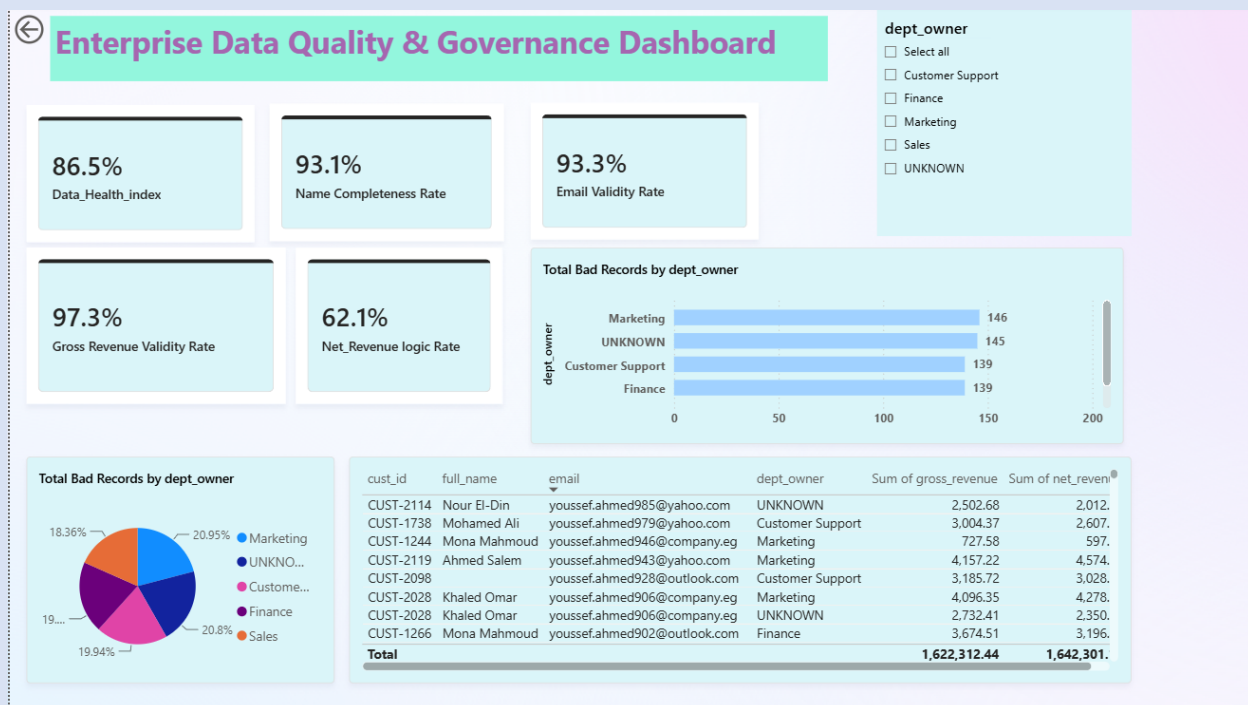
### 3. Cross-Field Data Mismatch (Consistency):

- **Issue:** Inconsistencies between customer names and associated email handles (e.g., mismatched identity records).
- **Remediation:** Assigned to Master Data Management (MDM) team for identity resolution.

### 4. Business Rule & Sign Inversion Errors (Accuracy):

- **Issue:** System logic errors causing negative gross values and inaccurate net revenue calculations.
- **Remediation:** Implemented DAX audit measures and flagged bad transactional rows.

## 5. Interactive Data Quality Dashboard (Power BI)



### Executive Summary of Key Metrics:

- **Overall Data Health Index: 86.5%**

- **Name Completeness Rate: 93.1%**
- **Email Validity Rate: 93.3%**
- **Gross Revenue Validity Rate: 97.3%**
- **Net Revenue Logic Rate: 62.1%**

**Operational Insight:** The primary driver of lower data quality scores stems from **Net Revenue Logic Errors (62.1%)**, predominantly concentrated within the **Marketing** and **UNKNOWN** departmental units. Remediation tasks have been logged in the Actionable Table visual for immediate Steward intervention.